

Food Bridge Application^{*,**}

Suhamam Mustafa^{a,1}, Aamir Yousaf^{a,2}, Muhammad Ali^{a,3}, Nazia Bibi^{a,*,5} and Shafa-at Ali Sheikh^{a,6}

^aDepartment of Computer Software Engineering, National University of Sciences and Technology, Islamabad, Pakistan

ARTICLE INFO

Keywords:

Mental Health
Sentiment Analysis
RoBERTa
Meme Generation
Personalized Recommendations
Roman Urdu
Gen AI

ABSTRACT

Food wastage is a rapidly increasing global challenge that negatively affects food security, economic efficiency, and environmental sustainability, despite the availability of surplus edible food. This study proposes a low cost, software based food redistribution platform designed to improve coordination between food donors and verified recipients through real time digital interaction and data driven decision support. The system enables users to post surplus food availability, request food, and monitor donations in real time using geographic matching and notification mechanisms. In addition to real time redistribution, historical food donation data is analyzed using a time-series forecasting approach based on the AutoRegressive Integrated Moving Average model to predict future surplus food trends. The forecasting results are visualized through time series and forecast graphs, as demonstrated in the system screenshots, supporting proactive planning and improved operational efficiency for food redistribution organizations. Experimental results show that integrating real-time digital coordination with predictive analysis reduces food spoilage, enhances transparency, and improves the overall effectiveness of food redistribution. The findings confirm that intelligent forecasting combined with digital platforms plays a critical role in reducing food wastage and strengthening sustainable community food systems..

1. Introduction

Food waste constitutes a growing global crisis with severe implications for food security, economic stability and environmental sustainability. Globally, 1.05 billion tonnes of food were wasted in 2022, at an economic cost of nearly \$1 trillion [1]. Paradoxically, 783 million people continue to face hunger worldwide [2]. Food waste alone accounts for 8-10% of global greenhouse gas emissions, demanding urgent reduction efforts [1]. The United Nations Sustainable Development Goal (SDG) 12.3 targets a 50% reduction in per capita food waste at the retail and consumer levels by 2030 [3].

Pakistan epitomizes these global challenges, underscoring the need for targeted interventions. The country loses approximately 19.6 million tonnes of food annually, around 26% of its total production [4]. This loss carries an economic cost of nearly \$4 billion [4]. Meanwhile, 43% of the population is food insecure and 18% face acute shortages [5]. Pakistan ranks 88th of 107 countries on the Global Hunger Index, with a score of 24.6 [6].

Overcoming this issue requires prevention of food waste instead of treating it after it occurred. Losses and waste occur at three different levels along the supply chain. Firstly, there are food production and post-harvest losses owing to lack of infrastructure and proper cold chain systems [7, 8]. Secondly, there is the distribution gap, during which edible excess food cannot reach the needy people [9]. Lastly, there are consumer level losses due to excessive purchase and poor storage systems [10, 1]. Among the three sources discussed, addressing the distribution gap emerges as the

most feasible preventive measure. It is more feasible because the surplus consists of edible food that can be redirected rather than wasted. Despite this, no reliable system currently exists in Pakistan to bridge this gap [11]. Proposed solutions have largely remained at the prototype or conceptual stage [11]. This paper therefore targets the distribution gap, where edible surplus food exists but no functional system connects it to those who need it.

Past studies have examined possible solutions in all three levels of food loss and wastage. At the production and post-harvest stage, AI tools have reduced food waste by up to 14.8% per store in commercial deployments [12]. However, these solutions target large-scale operations and remain unsuitable for small donors due to high infrastructure costs [8]. At the consumer level, mobile applications have improved food-sharing behaviour and reduced waste [9]. In relation to the distribution level, food donor-recipient platforms have been proposed. Most such platforms remain conceptual or limited in scale due to poor stakeholder reciprocity and weak platform performance [13, 14, 15]. These platforms have no capacity to forecast demand since they utilize manual/static matching [14, 15]. There have been no systems that integrate both food redistribution and demand prediction for Pakistan [11].

This research proposes a mobile platform connecting food donors, including individuals, households and restaurants with recipients such as NGOs and food banks. A history-based forecasting module enables recipients to anticipate incoming food supply, shifting redistribution from reactive to proactive. Both donor and recipient users register through dedicated accounts with no geographical restriction, ensuring accessibility across Pakistan. The rest of the paper is divided into sections as follows: Section 2 reviews the literature, Section 3 describes the methodology, Section 4

 suhamamustafa1@gmail.com (S. Mustafa);

ayousaf.msai24seecs@seecs.edu.pk (A. Yousaf); aliawan1170@gmail.com (M. Ali); nazia.bibi@mcs.nust.edu.pk (N. Bibi); ashafaat78@gmail.com (S.A. Sheikh)

ORCID(s):

presents the results, Section 5 discusses the findings, Section 6 concludes the paper and Section 7 outlines future work.

2. Related Work

This section reviews deployed food redistribution and food assistance platforms that are directly related to the proposed Food Bridge application. Existing solutions address surplus food redistribution through different operational models, including community-based food sharing, nonprofit food recovery networks, food assistance systems, business-to-charity redistribution, and technology-enabled traceability platforms [14, 16, 17, 18, 19]. These platforms differ in their target users, coordination mechanisms, and technological capabilities.

Community food-sharing applications, such as OLIO and Mehman Nawaz, mainly support peer-to-peer food exchange by allowing individuals to post or request food within nearby communities. In contrast, food recovery networks such as Rizq Bachao focus on collecting surplus food from restaurants, businesses, and partner organizations before redistributing it to vulnerable communities. Food assistance platforms, including Taqatwar Pakistan and Vaseela, primarily support ration distribution or donation-based food support rather than real-time surplus cooked-food redistribution. Technology-enabled platforms such as Regusto further introduce blockchain-based traceability, ESG reporting, and business-to-nonprofit redistribution.

Although these platforms contribute to food waste reduction and food access, several limitations remain. Most existing systems focus on only one stage of the redistribution process, such as posting, donation, recovery, or assistance. They generally lack an integrated workflow for donor registration, NGO coordination, real-time food claiming, volunteer assignment, pickup confirmation, donation lifecycle tracking, expiry monitoring, automatic re-listing, and history-based forecasting. These gaps motivate the development of the proposed Food Bridge application, which aims to provide an end-to-end digital ecosystem for surplus food redistribution.

Table 1 summarizes the reviewed platforms, their key features, and their limitations with respect to the proposed Food Bridge application.

2.1. OLIO

OLIO is a United Kingdom-based mobile food-sharing application that enables households, local businesses, and volunteers to redistribute surplus food within their local communities through a location-based platform [14]. The application allows users to post surplus food listings, browse nearby available food items, communicate with donors through in-app messaging, and arrange food collection. In addition, OLIO operates the Food Waste Hero program, where trained volunteers collect unsold food from local businesses and redistribute it within nearby communities. Harvey et al. investigated the effectiveness of mobile food-sharing platforms by analyzing 54,913 food-sharing exchanges among 9,054 users and concluded that platforms such as OLIO can effectively facilitate community-based

surplus food redistribution and reduce household food waste [14]. Figure 1 illustrates the OLIO mobile application interface used for community-based surplus food sharing.

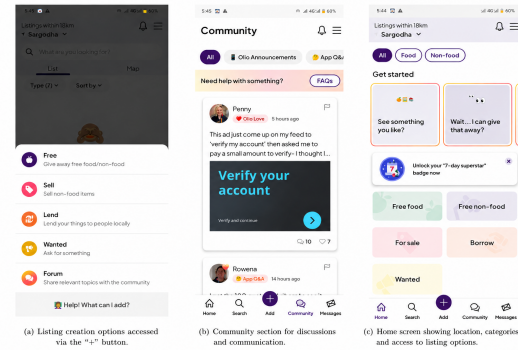


Figure 1: OLIO mobile application interface used for community-based surplus food sharing [14].

By comparing OLIO with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, OLIO does not provide history-based forecasting to estimate future surplus food availability.
- Unlike Food Bridge, OLIO does not support event-aware forecasting for occasions such as Ramadan, Eid, or weddings when surplus food generation typically increases.
- Unlike Food Bridge, OLIO does not provide complete donation lifecycle tracking with confirmed pickup status.
- Unlike Food Bridge, OLIO primarily supports peer-to-peer community food sharing rather than structured donor-NGO coordination and real-time food claiming workflow.

2.2. Mehman Nawaz

Mehman Nawaz is a Pakistan-based mobile food-sharing application that enables individuals, families, restaurants, and businesses to share surplus food with nearby users through a location-based platform [20]. The application allows users to create food listings by specifying the food quantity, category, photos, and pickup location. Users can browse nearby available food, communicate with donors through in-app chat, and discover food based on their current location. The platform aims to encourage community-based food sharing while reducing food waste through direct donor-recipient interaction. Figure 2 illustrates the Mehman Nawaz mobile application interface used for local food sharing.

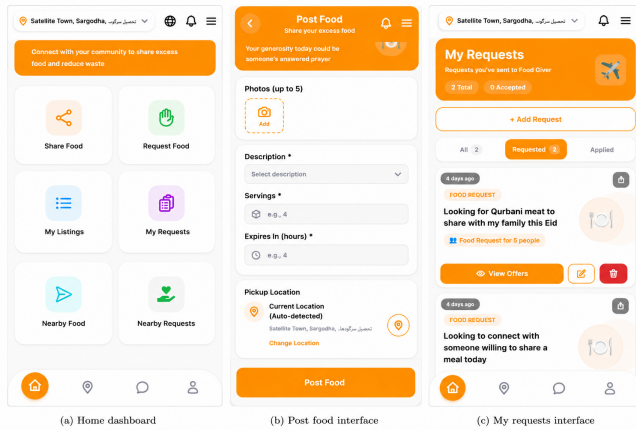


Figure 2: Mehman Nawaz mobile application interfaces showing (a) the home dashboard with food sharing and request options, (b) the food posting interface with photo upload, food details, expiry time, and pickup location, and (c) the request management interface for viewing submitted requests and available food offers [20].

By comparing Mehman Nawaz with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, Mehman Nawaz does not provide automatic forecasting of surplus food availability based on historical donation patterns.
- Unlike Food Bridge, Mehman Nawaz does not provide event-aware prediction for high-demand periods such as Ramadan, Eid, or weddings.
- Unlike Food Bridge, Mehman Nawaz does not provide complete donation lifecycle tracking, including confirmed pickups, donation history, cancelled requests, or re-posted food listings.
- Unlike Food Bridge, Mehman Nawaz does not automatically re-list unclaimed food, provide expiry countdowns, or remove expired food listings.
- Unlike Food Bridge, claim cancellation is not directly supported through the application workflow and generally requires manual communication between users.

2.3. Regusto

Regusto is an Italy-based digital redistribution platform that facilitates the donation and redistribution of surplus food and non-food products between companies and nonprofit organizations through a blockchain-enabled ecosystem [19]. The platform allows companies to donate surplus products, track donations through blockchain records, generate ESG impact reports, monitor waste reduction metrics, certify donation activities, and coordinate redistribution with nonprofit organizations. By integrating traceability and sustainability reporting, Regusto supports transparent and accountable redistribution of surplus resources. Figure 3 presents the Regusto digital redistribution platform.

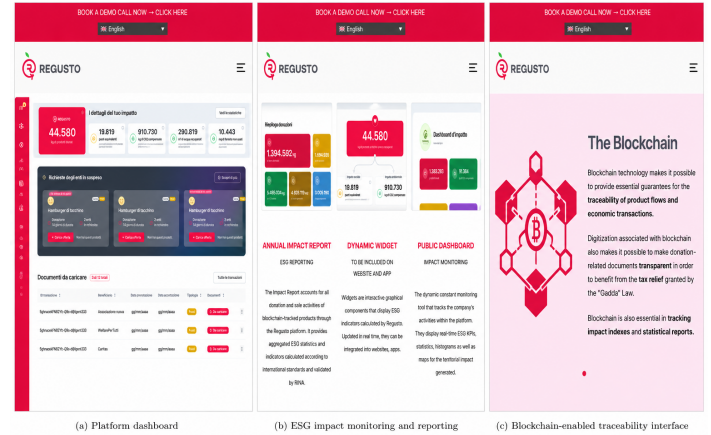


Figure 3: Regusto web platform interfaces supporting business-oriented surplus food redistribution and sustainability management [19].

By comparing Regusto with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, Regusto primarily targets businesses, restaurants, and nonprofit organizations rather than supporting ordinary individuals in donating surplus food.
- Unlike Food Bridge, Regusto is mainly a web-based platform and does not provide a mobile-first community food-sharing experience.
- Unlike Food Bridge, Regusto does not provide forecasting of future surplus food availability based on historical donation patterns.
- Unlike Food Bridge, Regusto does not support real-time browsing and claiming of surplus food by nearby NGOs or recipients.
- Unlike Food Bridge, some redistribution activities involve discounted sales or commercial transactions instead of completely free food donations.

2.4. Taqatwar Pakistan

Taqatwar Pakistan is a Pakistan-based food assistance platform developed to support food-insecure and underprivileged communities through a digitally coordinated ration distribution program [21, 22]. The platform allows donors and beneficiaries to register, submit beneficiary information for verification, apply for ration assistance, track application status, and receive food support through donor-funded initiatives. The primary objective of the platform is to facilitate organized food assistance by connecting donors with eligible beneficiaries through a structured verification process. Figure ?? shows the Taqatwar Pakistan platform used for ration assistance and donor coordination.

By comparing Taqatwar Pakistan with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, Taqatwar Pakistan focuses on ration assistance rather than surplus cooked-food redistribution by ordinary donors.
- Unlike Food Bridge, the platform does not allow users to create surplus food listings with quantity, category, photos, and pickup location.
- Unlike Food Bridge, nearby NGOs or recipients cannot browse and claim available surplus food in real time.
- Unlike Food Bridge, the platform does not provide forecasting of future surplus food availability based on historical donation patterns.
- Unlike Food Bridge, it does not provide complete donor-wise and recipient-wise surplus food tracking throughout the donation lifecycle.

2.5. Rizq Bachao

Rizq Bachao is a Pakistan-based food recovery and redistribution platform that works with businesses, restaurants, and charitable organizations to reduce food waste through resale, redistribution, and traceable recovery systems [23]. The platform enables businesses to donate surplus food, recover excess inventory, document donation activities, track impact metrics, and coordinate food recovery operations through Rizq's redistribution network. Its main focus is to convert surplus food into social, economic, and environmental value while improving food access for vulnerable communities. Figure 4 shows the Rizq Bachao food recovery and redistribution platform.

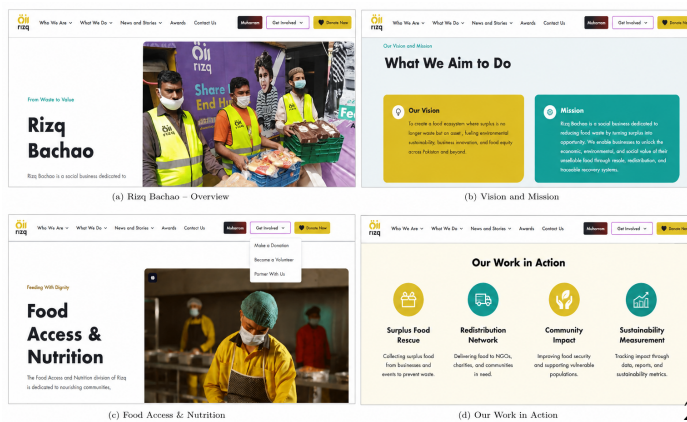


Figure: Rizq web platform interfaces supporting surplus food management, redistribution, and community nutrition initiatives.

Figure 4: rizqbachao [23].

By comparing Rizq Bachao with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, Rizq Bachao does not provide an open self-service platform where ordinary users can directly post surplus food.

- Unlike Food Bridge, recipients cannot browse and claim specific available food items in real time.
- Unlike Food Bridge, food collection and redistribution mainly depend on Rizq's operational team and partner network.
- Unlike Food Bridge, Rizq Bachao does not provide forecasting of future food availability based on historical donation patterns.
- Unlike Food Bridge, the platform does not provide a nationwide open mobile workflow for direct donor-NGO food claiming.

2.6. Too Good To Go

Too Good To Go is a Denmark-based mobile food waste reduction platform that enables restaurants, bakeries, supermarkets, and other food businesses to sell surplus food as discounted Surprise Bags through a mobile application [24, 14]. Users can browse nearby surplus food offers, reserve Surprise Bags, make in-app payments, and collect food from participating businesses within specified pickup windows. The platform helps reduce commercial food waste by redirecting unsold food to consumers at lower prices. Figure ?? shows the Too Good To Go mobile application interface.

By comparing Too Good To Go with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, Too Good To Go allows only registered businesses to list surplus food, while ordinary individuals cannot post food donations.
- Unlike Food Bridge, the platform works as a discounted food marketplace rather than a free donation-based redistribution system.
- Unlike Food Bridge, users usually do not know the exact contents of the Surprise Bag before purchase.
- Unlike Food Bridge, the platform does not provide NGO-based real-time food claiming and donor-recipient coordination.
- Unlike Food Bridge, the platform does not provide forecasting of future surplus food availability based on historical donation patterns.

2.7. Vaseela

Vaseela is a Pakistan-based food assistance platform that enables donors to financially support individuals and families experiencing food insecurity through a mobile application and digital donation system [25?]. The platform allows users to donate through the application, select donation amounts, support hunger relief initiatives, and contribute towards providing food assistance to underprivileged communities. It primarily serves as a digital channel for charitable giving and donor coordination to improve food accessibility across Pakistan. Figure ?? presents the Vaseela food assistance platform.

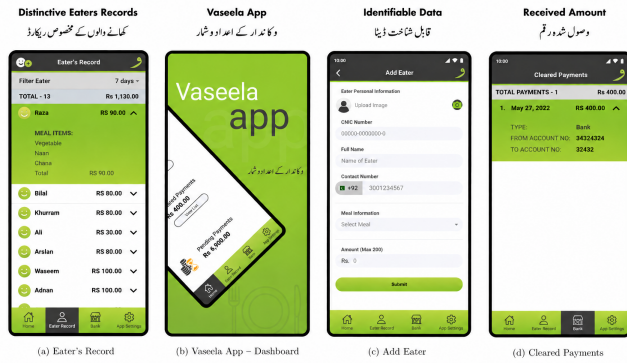


Figure 5: V

aseela mobile application interfaces showing (a) eater records management, (b) application dashboard, (c) eater registration interface, and (d) cleared payment records [25].

By comparing Vaseela with the proposed Food Bridge application, the following differences were identified:

- Unlike Food Bridge, Vaseela primarily focuses on food assistance through monetary donations rather than surplus cooked-food redistribution.
- Unlike Food Bridge, ordinary donors cannot create surplus food listings by specifying quantity, category, photos, and pickup location.
- Unlike Food Bridge, the platform does not provide real-time browsing and claiming of available surplus food by nearby NGOs or recipients.
- Unlike Food Bridge, the platform does not provide forecasting of future surplus food availability based on historical donation patterns.
- Unlike Food Bridge, the platform does not provide complete lifecycle tracking of surplus food donations from posting to successful distribution.

2.8. Summary of Related Work

To address these limitations, this research proposes the *Food Bridge* application, a mobile-based surplus food redistribution platform that integrates individuals, households, restaurants, NGOs, volunteers, and administrators within a unified digital ecosystem. Unlike existing platforms that address only individual stages of food redistribution, Food Bridge provides an end-to-end workflow including donor registration, NGO coordination, real-time food posting, automated recipient claiming, complete donation lifecycle tracking, expiry monitoring, donation history management, and centralized administrative control. In addition, the proposed system incorporates a history-based forecasting module to assist NGOs in anticipating future surplus food availability and planning food collection and distribution

activities more effectively. By integrating these functionalities within a single platform, Food Bridge is expected to improve operational efficiency, reduce food wastage, enhance transparency, and support proactive decision-making for NGOs and food banks, thereby addressing several limitations identified in existing food redistribution and food assistance platforms.

Table 1: Summary of Existing Studies

Digital Platform		Fields	Key Features	Limitation
OLIO [26]		Food Platform, Application, Community Redistribution	Sharing Mobile Users can post surplus food listings, request available food items, browse nearby food through geolocation, communicate with donors through in-app messaging, participate in Food Waste Hero collections, and share both food and household items within local household communities.	- Does not study past patterns to predict future surplus forecasting. - No event-based forecasting for seasonal demand fluctuations. - Doesn't confirm if posted items were picked up. - Ignores festivals or events that cause sudden surplus surges as no event-based prediction.
Mehman [20]	Nawaz	Food Platform, Application, Pakistan	Sharing Mobile Users can create food listings with quantity, category, photos, and pickup location, request nearby food, communicate through in-app chat, and use location-based food discovery.	-No automatic prediction of high-demand periods like Ramadan, Eid or weddings and no alerts based on past trends. -No confirmed pickup status, no detailed donation history and no record of cancelled or re-posted claims. - No auto re-listing if food isn't picked up, no expiry countdown or auto-removal. -Claim cancellation is unclear as there's no clear "Cancel" button and has to be done by manually messaging.
Regusto [27]		Digital Redistribution Platform, Blockchain, Italy	Companies can donate or redistribute surplus products, track donations through blockchain records, generate ESG impact reports, monitor waste reduction metrics, certify donation activities, and coordinate redistribution with nonprofit organizations through a digital platform	-Mostly web-based; not on Play Store in Pakistan. -No forecasting of future food surplus availability. -Designed mainly for businesses, restaurants and charities, not casual neighbor-to-neighbor sharing. Some listings involve discounted sales or payment not just free donations.
Taqatwar [21, 22]	Pakistan	Food Assistance Platform, Mobile Application, Pakistan	Users can register as donors or beneficiaries, submit beneficiary information for verification, apply for ration assistance, track application status, and receive food support through a digitally coordinated donor-funded assistance program.	- It focuses on ration assistance rather than surplus cooked-food posting by ordinary donors. - It does not allow nearby NGOs or recipients to browse and claim specific surplus food listings in real time. - It does not provide forecasting of future surplus food availability based on donation history. - It does not provide detailed donor-wise and recipient-wise surplus donation monitoring similar to Food Bridge.

Study	Fields	Key Features	Limitation
Rizq Bachao [23]	Food Recovery and Redistribution Platform, Pakistan	Businesses can donate surplus food through Rizq Bachao, food can be recovered and redistributed through partner organizations, donation activities can be documented, impact metrics can be tracked, and food recovery operations can be coordinated through Rizq's redistribution network	-An ordinary person with extra food can't simply open an app and post it themselves; pickup is done by Rizq's own team. -Only in cities/areas where Rizq has food banks or carts, unlike a nation-wide open app. -No forecasting of future food availability based on historical donation patterns. -Recipients don't browse and claim specific food items. Food is collected centrally then distributed.
Too Good To Go [24]	Food Waste Reduction Platform, Mobile Application, Denmark	Users can browse nearby surplus food offers, purchase discounted Surprise Bags, reserve food through the mobile application, make in-app payments, and collect food from participating restaurants, bakeries, and supermarkets during specified pickup windows.	-Only registered businesses can list food; regular people can't post or donate. - Fixed time windows with no easy reschedule; missing it can mean losing the food and the money paid. -Users pay for a "Surprise Bag" without knowing its exact contents before buying.
Vaseela [25, 28]	Food Assistance Platform, Mobile Application, Pakistan	Users can donate through the mobile application, select donation amounts, support people facing food insecurity, and contribute to food access for underprivileged communities. The platform is designed to connect donors with people in need and support hunger reduction in Pakistan.	- It focuses mainly on food assistance rather than surplus cooked-food redistribution. - It does not allow ordinary donors to post specific surplus food listings with quantity, photos, and pickup location. - It does not provide real-time NGO-based claiming of available surplus food. - It does not clearly provide donation-history-based forecasting or detailed surplus food monitoring similar to Food Bridge.

Table 2: Comparison of functional capabilities between the proposed Food Bridge application and existing food redistribution platforms.

Feature	Food Bridge	OLIO	Mehman Nawaz	Regusto	Taqatwar Pakistan	Rizq Bachao	Vaseela
Primary objective	Food redistribution	Community sharing	Community sharing	Business redistribution	Food assistance	Food recovery	Food assistance
Target users	Donors, NGOs, volunteers, admin	Individuals	Individuals	Businesses, nonprofits	Donors, beneficiaries	Businesses, NGOs	Donors, beneficiaries
Donor registration	Full	Full	Full	Full	Full	Full	Full
NGO registration	Full	None	None	Full	Partial	Full	Partial
Volunteer registration	Full	Partial	None	None	None	Partial	None
Role-based access control	Full	None	None	Partial	Partial	Partial	Partial
Surplus food posting	Full	Full	Full	Partial	None	Partial	None
Food photo upload	Full	Full	Full	Partial	None	None	None
Food category selection	Full	Partial	Full	Partial	None	Partial	None
Pickup location selection	Full	Full	Full	Partial	None	Partial	None
Real-time food posting	Full	Full	Partial	None	None	Partial	None
Real-time NGO claiming	Full	None	Partial	None	None	Partial	None
NGO claim management	Full	None	None	Partial	Partial	Full	Partial
Volunteer assignment	Full	Partial	None	None	None	Partial	None
Pickup confirmation	Full	Partial	None	Partial	None	Partial	None
Delivery confirmation	Full	None	None	Partial	None	Partial	None
Donation lifecycle tracking	Full	None	None	Partial	None	Partial	None
Donation history	Full	Partial	None	Full	Partial	Full	Partial
Food status tracking	Full	Partial	Partial	Full	Partial	Partial	None
Expiry countdown	Full	None	None	None	None	None	None
Automatic re-listing	Full	None	None	None	None	None	None
History-based forecasting	Full	None	None	None	None	None	None
Real-time synchronization	Full	Partial	Partial	None	None	None	None
Admin dashboard	Full	None	None	Full	Full	Full	Partial
Analytics dashboard	Full	None	None	Full	Partial	Partial	None
Donor verification	Full	None	None	Full	Partial	Partial	Partial
Multi-role architecture	Full	None	None	None	None	None	None
Role-specific dashboard	Full	None	None	Partial	Partial	Partial	Partial
Localization support	Full	Partial	None	None	Partial	None	Partial
Mobile application support	Full	Full	Full	None	Full	None	Full
Web application support	Full	Full	None	Full	Full	Full	Full
Community impact view	Full	Partial	None	Full	Partial	Full	Partial

Note: Full = feature completely supported; Partial = feature partially supported or limited implementation; None = feature not available.

References

- [1] United Nations Environment Programme. Food waste index report 2024: Think eat save — tracking progress to halve global food waste. Technical report, UNEP, Nairobi, Kenya, 2024.
- [2] Food and Agriculture Organization of the United Nations. The state of food security and nutrition in the world 2023. Technical report, FAO, Rome, Italy, 2023.
- [3] United Nations. Transforming our world: The 2030 agenda for sustainable development — sustainable development goal 12.3. Technical report, United Nations, New York, USA, 2015.
- [4] Ministry of National Food Security and Research. Report on food waste in pakistan. Technical report, Government of Pakistan, Islamabad, Pakistan, 2023.
- [5] World Food Programme. Pakistan food security report. Technical report, WFP, 2022.
- [6] Welthungerhilfe and Concern Worldwide. Global hunger index 2020: One decade to zero hunger. Technical report, Welthungerhilfe, 2020.
- [7] Blessing Iyanuoluwa Adediran, Akudo Francilia Onyegbula, Stephen Olufemi Oyeyipo, Tawakalitu Ahmed, Titilope Abosede Fashanu, and Damilola Olubunmi Ariyo. Smart postharvest management: Leveraging AI for reduced food loss, waste, and improved quality. *Journal of Agriculture and Rural Development Studies*, 2025(3):5–16, 2025.
- [8] M. Clark, S. Kanavikar, A. Clark, and C. Donnelly. Exploring the potential of ai-driven food waste management strategies used in the hospitality industry for application in household settings. *Frontiers in Sustainable Food Systems*, 9, 2025.
- [9] C. Agmapisarn, P. Seetha, and P. Fakfare. Optimizing surplus food redistribution: Leveraging digital platforms to advance sustainable consumption. *Sustainable Development*, 33:3831–3846, 2025.
- [10] V. Aleksanyan, S. Gevorgyan, D. Markosyan, F. H. Arion, K. Khachatryan, F. C. Oroian, I. C. Muresan, I. D. Arion, and S. Chis. Managing food waste in the restaurant sector: Comparative insights from greece and armenia. *Sustainability*, 17(24):11386, 2025.
- [11] M. Nisar et al. Usability evaluation of food wastage mobile applications in pakistan. *International Journal of Advanced Computer Science and Applications*, 2021.
- [12] R. Cejas. Food waste reduction ai technologies in restaurant management: An ms-toro approach. *Processes*, 2025.
- [13] F. de Almeida Oroski and J. M. da Silva. Understanding food waste-reducing platforms: A mini-review. *Waste Management Research*, 41(4):816–827, 2023.
- [14] J. Harvey, A. Smith, J. Goulding, and I. Branco Illodo. Food sharing, redistribution, and waste reduction via mobile applications: A social network analysis. *Industrial Marketing Management*, 88:437–448, 2020.
- [15] Alice Mazzucchelli, Martina Gurioli, Domenico Graziano, Barbara Quacquarelli, and Chiraz Aouina-Mejri. How to fight against food waste in the digital era: Key factors for a successful food sharing platform. *Journal of Business Research*, 124:47–58, 2021.
- [16] T. Kenny and C. Sage. Surplus food redistribution and healthy, sustainable diets: Exploring the contradictions of charitable food provisioning. *International Journal of Sociology of Agriculture and Food*, 27(1):71–86, 2021.
- [17] A. Parthasarathy. Tackling food insecurity through food surplus redistribution. *American Journal of Student Research*, 3(5):93–102, 2025.
- [18] A. Veselá. Surplus food redistribution: Towards a world where no good food goes to waste. Master's thesis, Utrecht University, 2022.
- [19] M. Yu, L. Principato, M. Formentini, et al. Unlocking the potential of surplus food: A blockchain approach to enhance equitable distribution and address food insecurity in italy. *Socio-Economic Planning Sciences*, 93:101868, 2024.
- [20] Mehman nawaz mobile application. <https://play.google.com>. Accessed: 2026.
- [21] Sunridge taqatwar pakistan. <https://taqatwarpakistan.org/>. Accessed: 2026.
- [22] Taqatwar pakistan mobile application. <https://play.google.com/store/apps/details?id=com.taqatwarpakistan>. Accessed: 2026.
- [23] Rizq bachao. <https://rizq.org>. Accessed: 2026.
- [24] Too good to go. <https://www.toogoodtogo.com>. Accessed: 2026.
- [25] Vaseela food assistance platform. <https://vaseela.net/>. Accessed: 2026.
- [26] Olío food sharing app. <https://olioapp.com>. Accessed: 2026.
- [27] Regusto platform. <https://www.regusto.eu>. Accessed: 2026.
- [28] Vaseela mobile application. <https://play.google.com/store/apps/details?id=io.insignia.vaseela>. Accessed: 2026.